🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.6C

Write and Evaluate Numerical Expressions with Exponents

Lesson 6-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.

Language Objective: I can explain my steps using the words expression, evaluate, order of operations, base, and exponent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Write and Evaluate Numerical Expressions with Exponents | Numerical expression | Evaluate | Order of operations | Power | Base | Exponent

👆 *Tap any word to see what it means and a picture.*

1. Write the situation as an expression, then follow the ________ ________ ________ to value it.

2. Numbers and operations with no equal sign and no variable form a ________ ________.

3. To put numbers in for the letters and work out the value is to ________.

4. Grouping symbols, then ________, then × and ÷, then + and −.

5. A number written with a base and an exponent, like 2^3, is a ____________.

6. In 2^3, the number used as a factor is the ____________.

7. The small raised number saying how many times the base is a factor is the ____________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How much will the ride cost? How does the expression model this situation?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Write and Evaluate Numerical Expressions with Exponents** — Turning a situation into a number sentence without an equal sign, then finding its value in the correct order.
  Escribir y evaluar expresiones numéricas con exponentes — Convertir una situación en una expresión numérica sin signo de igual y luego hallar su valor en el orden correcto.
- [ ] **Numerical expression** — A math phrase built from numbers and operations, with no equal sign and no variable.
  Expresión numérica — Una frase matemática hecha de números y operaciones, sin signo de igual y sin variable.
- [ ] **Evaluate** — To find the single value an expression is worth by carrying out its operations in order.
  Evaluar — Hallar el único valor que representa una expresión, realizando sus operaciones en orden.
- [ ] **Order of operations** — The agreed order for evaluating: grouping symbols, then exponents, then multiply and divide left to right, then add and subtract left to right.
  Orden de las operaciones — El orden acordado para evaluar: signos de agrupación, luego exponentes, luego multiplicar y dividir de izquierda a derecha, y por último sumar y restar de izquierda a derecha.
- [ ] **Power** — A short way to write repeated multiplication of the same factor, such as 2^3 for $2 \times 2 \times 2$.
  Potencia — Una forma corta de escribir la multiplicación repetida del mismo factor, como 2^3 para $2 \times 2 \times 2$.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The ride costs $____ because 2.50 + 1.75(____) = ____. The 2.50 is the ____ and 1.75 is the ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Substitute the number of speakers for s, then evaluate.

Evidence: Students substitute $s = 5$ ($150 + 12 \times 5 = 210$), explain 150 is the fixed venue fee and 12 is the cost per speaker, and note s can be any number.

Reasoning: This evidence connects the definition of write and evaluate numerical expressions with exponents to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The ride costs $____ because 2.50 + 1.75(____) = ____. The 2.50 is the ____ and 1.75 is the ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Write and Evaluate Numerical Expressions with Exponents
2. **Notes 2:** Numerical expression
3. **Notes 3:** Evaluate
4. **Notes 4:** Order of operations
5. **Notes 5:** Power
6. **Notes 6:** Base
7. **Notes 7:** Exponent
8. **Watch for this mistake:** *A common mistake with powers is multiplying the base by the exponent: reading 3^2 as $3 \times 2 = 6$ instead of $3 \times 3 = 9$. A second is evaluating strictly left to right, so $3 + 4 \times 2$ becomes 14 instead of 11. The order of operations exists so one expression has exactly one value: grouping symbols, then powers, then × and ÷ left to right, then + and − left to right.*
9. **Try It 1:** *— Grouping symbols outrank everything; powers outrank multiplying; multiplying outranks adding and subtracting.*
10. **Try It 2:** *— Each power is repeated multiplication of the base, as many times as the exponent says.*